



Project Scope and Plan - AI Factories

Access Call for Science and Collaborative EU Projects

Project title	DeltaTok: Efficient Geometry-Grounded 4D World Models with Flexible Camera Setups
Research field	PE6 - Computer Science and Informatics (computer vision, machine learning, autonomous driving)
AI technologies used	Vision transformers; flow-matching generative models (4D world models); latent tokenizers / representation autoencoders; knowledge distillation from 2D foundation models; self-supervised learning; distributed training (PyTorch DDP/FSDP, NCCL, BF16, FlashAttention-2, WebDataset)
Proposal abstract	Generative world models are becoming the simulation backbone of autonomous driving (AD), yet current video-based models (GAIA-2, Cosmos, Vista) hallucinate geometry, drift over time, and are locked to the fixed, calibrated camera rig they were trained on. Building on our urban 3D geometry foundation model OccAny (CVPR 2026), we propose DeltaTok, an efficient geometry-grounded 4D world model that natively supports flexible camera setups: monocular, surround-view, and arbitrary uncalibrated rigs. Instead of generating pixels, DeltaTok tokenizes the intermediate feature space of the frozen OccAny geometry encoder into a compact, geometry-grounded latent code (DeltaTok surround), and trains a flow-matching transformer (DeltaTok-Flow) to predict future scene tokens conditioned on past observations and the ego trajectory. Predicted tokens are decoded into metric depth, point maps, camera poses and 3D occupancy through the frozen OccAny decoder, and into RGB through a lightweight MAE-style image decoder — yielding a physically consistent neural simulator at a fraction of the cost of pixel-space video diffusion. All pipelines are already deployed and benchmarked on MareNostrum 5 (ehpc793, fast-lane EHPC-AIF-2026FL01-008); the requested allocation funds the tokenizer/world-model development and scale-up. All code and checkpoints will be released open-source.



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Co-PIs/Team members (same information)

Provide the details of any Co-PIs and Team members in the project, including Title, First (given) name, Last (family) name, Organisation, Department, Group and Country.

It is mandatory to include all team members on the online form.

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1 EXCELLENCE (~ 3 pages)

1.1 Proposed methodologies and outcomes

Concept and motivation. World models — generative models that predict how a driving scene evolves conditioned on the ego vehicle's actions — are emerging as the key enabling technology for closed-loop simulation, safety validation and synthetic data generation in autonomous driving (AD). The dominant approach generates future camera frames directly in pixel space with large video diffusion/flow models (GAIA-2 [4], Cosmos [5], Vista [6], GEM [7]). This paradigm has three structural weaknesses. First, it is computationally inefficient: most of the modelling capacity is spent re-synthesising appearance (textures, lighting) rather than the scene dynamics that matter for driving. Second, it is not physically grounded: purely 2D objectives produce geometric flickering, inconsistent depth and implausible object interactions, which disqualify the outputs for safety-critical simulation. Third, it is rig-locked: models are trained for one fixed, calibrated camera configuration and do not transfer to a different number or placement of cameras.

Our approach. We address all three weaknesses by moving generation from pixel space into the latent space of a geometry foundation model. Our recent model OccAny [1] (CVPR 2026, built on Depth Anything 3 [3], 1.1B parameters) is the first unconstrained urban 3D occupancy model: from monocular, sequential or surround-view images — without calibration — it jointly predicts metric point maps, depth, camera poses and semantic 3D occupancy. Its internal representation is therefore a compact, geometry-grounded, camera-flexible description of the scene: exactly the state a world model should evolve. We have verified experimentally (token-caching round-trip at encoder layer 18 of the frozen OccAny backbone) that this intermediate representation preserves the full reconstruction quality of the model, halving the cached feature size by exploiting the local-attention structure of the backbone.

Objectives. The project develops and scales, on MareNostrum 5, the two components that turn OccAny into an efficient 4D world model:

- **WP1 — DeltaTok surround tokenizer.** A tokenizer trained over per-frame OccAny layer-18 features that compresses them into a small set of discrete/continuous latent tokens per timestep. Resolution-flexible positional embeddings (learned 2D grid with bicubic interpolation + 2D RoPE, following the DA3 design) make the same tokenizer applicable to any camera count and image resolution — the mechanism behind flexible camera setups. Temporal delta-encoding exploits the redundancy between consecutive timesteps of a driving scene.
- **WP2 — DeltaTok-Flow world model.** A flow-matching transformer [8] over token sequences that predicts the scene state at future timesteps, conditioned on past tokens and the ego trajectory through cross-attention. Factorised spatial (within-view) and temporal/cross-view attention keeps the cost linear in the number of cameras.



Logit-normal timestep sampling and v-prediction follow current best practice for flow models.

- **WP3 — Decoders.** Predicted tokens are decoded two ways: (i) geometrically, by resuming the frozen OccAny backbone from layer 19, yielding metric depth, point maps, poses and 3D occupancy for free — every generated future is natively annotated in 3D; and (ii) photometrically, with a lightweight MAE-style image decoder [12] trained on frozen multi-level OccAny features, producing RGB for visual inspection and FVD-style evaluation.

- **WP4 — Scaling and evaluation.** Joint scaling of model capacity (1.3B → 3B), temporal horizon and data volume (nuScenes, Waymo, DDAD, PandaSet, plus the 5,500-hour multi-camera NATIX corpus [15]), with continuous geometric evaluation: occupancy IoU, depth/point-map accuracy, and trajectory consistency (ADE between trajectories estimated from real vs. generated sequences). Our in-training evaluation harness (200-sample nuScenes Vista protocol, CVPR-baseline mean ADE 0.95) is already operational.

Choice of computational methods. All components are dense transformers trained with PyTorch DDP (FSDP for the largest configurations), BF16, FlashAttention-2 and gradient accumulation; data are streamed as sharded WebDataset archives directly from GPFS — a pipeline we hardened during the ehpc793 fast-lane (see Section 3.1). Flow matching is preferred over diffusion for its straighter probability paths and fewer sampling steps, directly reducing both training and inference cost. Operating in token space reduces the sequence length per frame by more than an order of magnitude compared to pixel-space latents of video VAEs, which is what makes systematic architecture search at 1.3B scale affordable [TODO: confirm exact token-count reduction factor from final DeltaTok config].

Main advances and outcomes. (1) The first geometry-grounded world model that is camera-setup-agnostic by construction; (2) a public, reusable geometry-token interface to a 3D foundation model (tokenizer + decoders); (3) scaling evidence for token-space world models on European HPC; (4) open-source code, training recipes and checkpoints for all components, released under permissive licences as with our previous works (MonoScene, PaSCo, OccAny).

1.2 Resources request justification

Why this requires EuroHPC scale. The computational demand is driven by (i) the 1.1B-parameter frozen geometry encoder through which every training sample passes, (ii) multi-camera, multi-timestep inputs (5–6 cameras × up to 25 timesteps per clip), and (iii) the breadth of the design space that must be explored for a new class of model (tokenizer rate/distortion trade-offs, conditioning strategies, attention factorisations, horizon curricula). Our cost anchors are measured, not estimated: on MareNostrum 5 ACC (fast-lane EHPC-AIF-2026FL01-008, project ehpc793) one production training run of a 1.3B flow-matching model — 8 nodes × 4 H100, batch size 1 per GPU, 300,000 iterations,



48 hours — costs exactly 1,536 H100 GPU-hours (384 node-hours). OccAny+ 1.1B geometry-encoder training runs at ~ 0.9 s/iteration on the same 32-GPU configuration (200,000 iterations ≈ 50 h). The milestones table in Section 3.2 builds every line item from these measured rates; the total request is 12,500 node-hours (50,000 H100 GPU-hours) [TODO: confirm total request matches the call ceiling].

Why it cannot run on our local cluster. Our home institution (valeo.ai) operates a small shared GPU cluster sized for inference and single-node fine-tuning; sustained multi-node H100 training is not possible there: a single WP4 scale-up run (8 nodes $\times$ 48 h) would monopolise more H100s than the entire cluster provides for two days, and the full programme (~ 75 multi-GPU training runs over 12 months) exceeds local annual capacity by more than an order of magnitude [TODO: confirm local cluster size statement]. Other allocations we hold are committed to the OccAny perception line and are capacity-limited for this generative workload. MareNostrum 5 is the only system where (a) we have already ported, benchmarked and hardened this exact code base, (b) 32–64 GPU jobs schedule with reasonable turnaround, and (c) the GPFS + node-local storage hierarchy matches our streaming WebDataset pipeline.

Readiness. Under ehpc793 we have consumed $\sim 18,600$ of 40,000 GPU-hours productively: the full training stack (launchers, fault-tolerant requeueing, checkpointing every 2,500 iterations, streaming I/O) is deployed and validated on MN5; six distinct multi-node failure modes (GPFS demand-paging stalls, NCCL straggler timeouts, node-local cache exhaustion, dataloader RAM growth) were diagnosed, fixed and documented, and the resulting configuration now sustains 1.4 it/s over multi-day 32-GPU runs with automatic requeue. The requested allocation therefore converts directly into science, not into porting effort.

2 INNOVATION AND IMPACT (~ 3 pages)

2.1 Innovative aspects and foreseen impact

Innovative aspects. The central innovation is the change of generative substrate: we generate the future in the latent space of a 3D geometry foundation model rather than in pixel or video-VAE space. This single design decision yields three properties no existing AD world model combines:

- **Geometry-grounded by construction.** Every generated token sequence decodes — through the frozen OccAny decoder — into metric depth, point maps, camera poses and semantic 3D occupancy. Physical consistency is not a soft auxiliary loss on rendered pixels; it is the native output format. Generated futures come densely annotated in 3D at zero additional cost, which directly addresses the long-tail labelling bottleneck in AD.
- **Flexible camera setups.** OccAny is trained on unconstrained, uncalibrated monocular/sequence/surround inputs, and the DeltaTok tokenizer inherits this through



resolution-flexible positional embeddings and per-view tokenisation. The same world model can therefore ingest and predict for different vehicle platforms (different camera counts, placements, resolutions) without retraining — zero-shot sensor-rig generalisation, where competing models are locked to one rig.

– **Efficiency.** Token-space generation shrinks the transformer sequence length per frame by over an order of magnitude versus pixel-space latents, and flow matching needs far fewer sampling steps than diffusion. Training cost per experiment drops accordingly (measured: 1,536 GPU-h per 300k-iteration 1.3B run), enabling the systematic exploration that pixel-space models cannot afford; at inference, short-horizon rollouts target interactive rates for closed-loop use [TODO: confirm inference latency target once first DeltaTok-Flow checkpoints exist].

Technological impact. The project shifts the paradigm from “video generation that looks right” to “scene prediction that is measurably right”. We target trajectory-consistency improvements of over 20% versus pixel-space baselines (Vista, GEM) measured by displacement error between trajectories estimated from real vs. generated sequences, at comparable visual quality (FID/FVD on nuScenes). Because outputs are metric and 3D, downstream consumers (planners, simulators) can use them directly — a capability gap no current open model fills.

Economic and industrial impact. For the European automotive sector, the framework attacks the validation bottleneck: critical edge cases move from physical road testing to compute-driven generation. Geometry-grounded generation natively provides the 3D annotations whose manual creation dominates dataset cost, and rig-flexibility lets OEMs re-render existing fleets' data for new camera configurations instead of re-driving millions of kilometres. We aim for generated data of sufficient quality to be mixed with real data at a 20–30% ratio in perception training. The resulting models will be, to our knowledge, the first open-source geometry-grounded multi-camera world model trained at this scale; based on the adoption of comparable open models (NVIDIA Cosmos: ~36,000 monthly downloads), we conservatively target 10,000 downloads within six months of release.

Societal impact. Rigorous virtual testing of rare, dangerous scenarios (occlusions, vulnerable road users, near-collisions) supports the EU “Vision Zero” road-safety goal: we expect to expose planners to 10–100× more critical scenarios per day than physical fleet testing. Shifting validation mileage into energy-efficient EuroHPC simulation also cuts the carbon footprint of AD development. Open-sourcing the models counters the concentration of world-model capability in a handful of non-EU industrial labs and keeps European academia and SMEs at the state of the art.

2.2 State-of-the-art and comparison with state-of-the-art

Pixel-space driving world models. Vista [6] (NeurIPS 2024), GEM [7] (CVPR 2025), GAIA-2 [4] and Cosmos [5] generate impressive video but share the weaknesses our proposal targets: geometry emerges only implicitly, temporal consistency degrades over long horizons, outputs carry no metric 3D information, and each model is bound to its



training rig. Recent geometry-aware variants (GeoDrive, Geometry Forcing) add depth or geometry losses on top of pixel generation, confirming the community's diagnosis — but they retrofit geometry as a regulariser rather than adopting a geometric state space, and remain single-rig and computationally heavy.

Occupancy/LiDAR world models. Token-space prediction over occupancy grids (e.g. OccWorld-style approaches) is geometric but depends on calibrated rigs and LiDAR-derived voxel labels [11], inherits the coarse resolution of the voxel grid, and produces no photometric output for visual validation. Our token space is learned from images by a foundation model trained on unconstrained data, retains photometric decodability, and is rig-agnostic.

Geometry foundation models. DUS3R [2], MonST3R and Depth Anything 3 [3] established that large-scale point-map regression generalises across domains, and OccAny [1] extended this line to complete semantic 3D occupancy for urban scenes — surpassing visual-geometry baselines on occupancy prediction while remaining competitive with in-domain self-supervised methods across monocular, sequential and surround inputs on two urban benchmarks. None of these models, however, predicts the future; they are perception models. Our project is, to our knowledge, the first to use such a model's latent space as the state space of a generative world model.

Honest positioning. We do not expect to match the raw visual fidelity of frontier pixel-space models trained on orders of magnitude more compute (GAIA-2, Cosmos are industrial efforts at 100× our budget); our RGB decoder targets evaluation-grade, not cinema-grade, imagery. Conversely, on the axes that matter for simulation — metric accuracy, temporal/geometric consistency, 3D annotation completeness, cross-rig transfer, and cost per generated scenario — we expect to surpass them, and we will report both axes transparently. The main scientific risks are (i) that layer-18 tokens prove insufficient for long-horizon dynamics (mitigation: multi-level tokenisation over the 3-level feature pyramid already used by our image decoder) and (ii) tokenizer/world-model error compounding (mitigation: joint fine-tuning in WP4).

3 QUALITY AND EFFICIENCY OF THE IMPLEMENTATION (~ 4 pages)

3.1 Technical feasibility of the proposal

Codes and software stack. The project uses two code bases, both already deployed on MareNostrum 5 under ehpc793: (1) OccAny (public: github.com/valeoai/OccAny) — the geometry foundation model, training, inference and evaluation pipelines (PI's work); (2) DeltaTok (this project; to be open-sourced) — tokenizer (train_deltatok), flow world model (train_deltatok_flow), MAE-style image decoder (train_occrac_img_decoder) and dataset-wide feature extraction, all Hydra-configured with SLURM-native distributed launchers. Stack: Python 3.12, PyTorch 2.6 + CUDA 12.6, xFormers/FlashAttention-2, NCCL, torchrun/srun, WebDataset, TensorBoard. Third-party components (Depth Anything 3,



SAM3, DUST3R/CroCo) are vendored for reproducibility. Mixed-precision BF16 throughout; checkpointing every 2,500 iterations with automatic SLURM requeue (validated in production).

GPU efficiency and scaling (measured on MN5 ACC, ehpc793). Figure 1 shows strong and weak scaling of the 1.3B flow-matching training measured on MareNostrum 5 H100 nodes under fast-lane EHPC-AIF-2026FL01-008. Strong scaling (fixed global batch 8): 2.62 s/batch on 1 GPU $\rightarrow$ 1.50 s on 2 GPUs (87.6% efficiency) $\rightarrow$ 0.82 s on 4 GPUs (80.3%). Weak scaling (batch $\propto$ GPUs): 90.3% efficiency at 2 GPUs and 89.5% at 4 GPUs — our primary operating regime. At production scale (8 nodes / 32 H100, DDP), the same code sustains 1.4 it/s over multi-day runs, i.e. 300,000 iterations in 48 h = 1,536 GPU-hours per run; memory constraints (1 clip per GPU) are handled with gradient accumulation. Extrapolating the measured weak-scaling curve we conservatively project 60–70% efficiency at 64 GPUs for the WP4 scale-up runs.

I/O and robustness on MN5. Training data are packed as sharded WebDataset tar archives and streamed directly from GPFS with overlapped reads (no node-local cache — we measured a $\sim$ 0% cache hit rate for resampled streaming and removed it), bounded prefetch buffers, and 120 s read timeouts. During the fast-lane we diagnosed and permanently fixed six multi-node failure modes (GPFS demand-paging stalls from /gpfs/home, NCCL straggler watchdog timeouts via broadcast_buffers=False, node-local /scratch exhaustion, cache-thrash slowdowns, dataloader host-RAM growth); the hardened configuration has since completed multiple uninterrupted 32-GPU production runs. Aggregate read bandwidth requirement is modest ($\sim$ 10 MB/s/GPU); checkpoints are a few GB each.

Feasibility of the plan. Every run type in the milestones table is either already running today at smaller scale (the tokenizer, flow and decoder trainers already run in production at 16 GPUs; OccAny+ 1.1B training and the full evaluation harness run on MareNostrum 5) or is a measured configuration. No new porting work is required.

1.3B flow-matching world model — measured scaling on MareNostrum 5 ACC (project ehpc793, fast-lane EHPC-AIF-2026FL01-008)

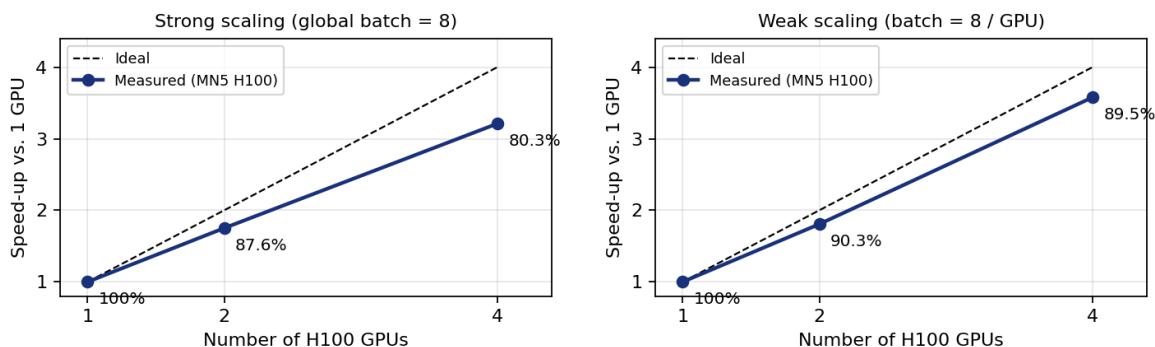


Figure 1: Measured strong/weak scaling of the 1.3B flow-matching training on MareNostrum 5 ACC H100 nodes (project ehpc793, fast-lane EHPC-AIF-2026FL01-008).



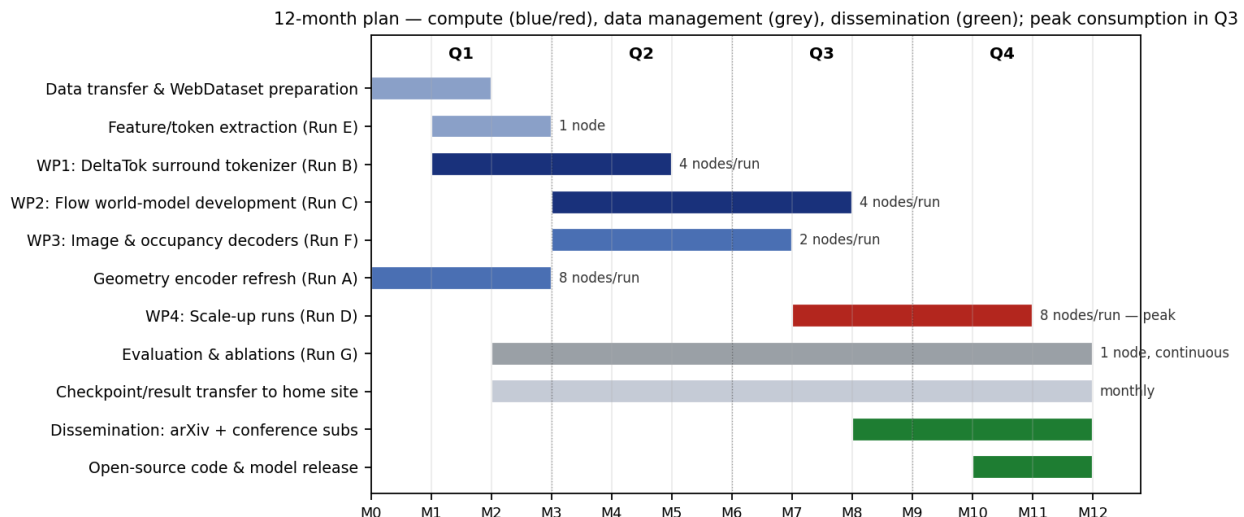
3.2 Milestones (quarterly basis)

Run Type	Code(s)	No. of runs	No. of nodes or GPUs	No. of steps per run	No of cores per node	Time per step(s)	Total node hours
A – Geometry encoder refresh	OccAny +	6	8 nodes (32 GPU)	200,000	80	0.90 (measured)	2,400
B – Surround tokenizer dev.	DeltaTok	20	4 nodes (16 GPU)	100,000	80	0.80 [TODO]	1,780
C – World-model dev. (1.3B)	DeltaTok-Flow	30	4 nodes (16 GPU)	100,000	80	1.00 [TODO]	3,333
D – Scale-up ($\leq 3B$, long horizon)	DeltaTok-Flow	8	8 nodes (32 GPU)	300,000	80	0.58 (measured)	3,093
E – Feature/token extraction	OccAny + enc.	12	1 node (4 GPU)	— (24 h/run)	80	—	288
F – Image & occ. decoders	DeltaTok dec.	12	2 nodes (8 GPU)	100,000	80	0.60 [TODO]	400
G – Evaluation & ablations	OccAny +/ DeltaTok	300	1 node (4 GPU)	— (≈ 1 h/run)	80	—	300
TOTAL ($\approx +8\%$ contingency $\rightarrow$ request 12,500)							11,594

All node-hours are computed as runs $\times$ nodes $\times$ steps $\times$ time-per-step / 3600 from rates measured on MareNostrum 5 ACC (rows A, D) or from current production configurations on comparable H100 systems (rows B, C, F; conservatively rounded — [TODO: confirm B/C/F time-per-step on MN5 with one calibration run each]). Rows E and G are wall-clock-based inference workloads (24 h and 1 h per run respectively). Quarterly commitments: Q1 (M1–M3): data staging, run types E + A and first tokenizer runs (B) — $\approx 2,900$ node-h. Q2 (M4–M6): complete B, main world-model development C and decoders F — $\approx 3,600$ node-h. Q3 (M7–M9): finish C, scale-up runs D — $\approx 4,100$ node-h (consumption peak). Q4 (M10–M12): remaining D, final evaluations G, release — $\approx 1,900$ node-h. Total: 11,594 node-h $\approx 12,500$ node-h requested including $\sim 8\%$ contingency for failed/repeated runs (justified by the multi-node failure statistics of our fast-lane) [TODO: confirm node-hour totals and contingency policy against the call rules].



3.3 Gantt Chart



Data transfer plan. Q1: training corpora (nuScenes, Waymo, DDAD, PandaSet subsets repacked as WebDataset shards, plus NATIX subsets as released) are staged to /gpfs/projects and /gpfs/scratch via the MN5 transfer nodes — ≈ 30 TB inbound [TODO: confirm dataset volume after shard packing]. Layer-18 feature/token dumps (run type E) are produced on-system (≈ 20 TB on scratch, recyclable) [TODO: confirm token dump size from extraction config]. Outbound: selected checkpoints and evaluation artefacts, ≈ 2 TB/month, transferred monthly to the home site and mirrored to Hugging Face on release; only metric-best checkpoints are retained long-term. Storage request: 60 TB scratch / 10 TB project [TODO: confirm storage request fields in the online form].

Communication and dissemination plan. Results are submitted to top-tier venues (CVPR/ICCV/NeurIPS) at the Q3/Q4 boundaries with immediate arXiv preprints; code is released on GitHub (valeoai organisation, as for OccAny/MonoScene/PaSCo) and checkpoints on Hugging Face; we present at European AD/CV workshops and to the EuroHPC user community. EuroHPC JU and BSC/MareNostrum 5 will be credited in all publications and model cards, and the allocation recorded in dataset/model provenance. The work aligns with the EU projects ELSA and ELLIOT in which the co-PI participates, providing additional dissemination channels.

3.4 Personnel and Management Plan

The core team comprises Dr. Anh-Quan Cao (PI, Research Scientist, valeo.ai Paris) and Dr. Tuan-Hung Vu (co-PI, Senior Research Scientist, valeo.ai Paris), who have collaborated for several years (MonoScene, OccAny) and co-authored the OccAny foundation model on which this project builds. Both are already in place; no hiring is required for the core programme. Anh-Quan Cao leads the technical work: DeltaTok tokenizer and flow world model development, MN5 operations (he ran the ehpc793 fast-lane, including the



distributed-training hardening described in Section 3.1), and evaluation. Tuan-Hung Vu guides the research strategy, leads data acquisition/preparation (Waymo, DDAD, PandaSet, NATIX) and the dissemination work package, and supervises the comparison against pixel-space baselines. Management is lightweight and proven: weekly technical reviews, a shared experiment log linking every SLURM job to observations and decisions (operational since the fast-lane), and quarterly checkpoint reviews aligned with the milestone table. The project can draw on additional valeo.ai engineering support (e.g. Dr. David Hurych, EXA4MIND) for data engineering peaks [TODO: confirm supporting personnel availability].



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